CHAPTER 3

Limits of functions

S21, S23, S24, S26

Objectives

Calculate limits at a point and at infinity; resolve an indeterminate form; interpret an asymptote.

Course notes

Meaning of a limit

$f(x) \to \ell$ as $x \to +\infty$ means that $f(x)$ can be made as close as desired to ℓ for all sufficiently large x. For a limit at a, we consider values of x in the domain, close to a and different from a. The left-hand and right-hand limits are written $\lim_{x\to a^-} f(x)$ and $\lim_{x\to a^+} f(x)$. When the domain has points on both sides of a, a two-sided limit exists if and only if both one-sided limits exist and are equal. At an endpoint of the domain, we study the limit from the only accessible side.

Graphical interpretation

If $f(x) \to \ell \in \mathbb{R}$ as $x \to +\infty$ or $-\infty$, the line $y = \ell$ is a horizontal asymptote in that direction. If $f(x) \to \pm\infty$ as $x \to a$ from one side, the line $x = a$ is a vertical asymptote.

A curve can cross a horizontal asymptote. The limit does not necessarily give the position of the curve relative to the asymptote: study the sign of $f(x) - \ell$.

Reference limits and operations

At $+\infty$, $x^m \to +\infty$ for $m \geq 1$, $\frac{1}{x^m} \to 0$, $\sqrt{x} \to +\infty$. At $-\infty$, the sign of x^m depends on the parity of m. The rules for sum, product, quotient and comparison are those for sequences, with the same indeterminate forms.

Example. For $f(x) = \frac{2x+1}{x-3}$, write $f(x) = 2 + \frac{7}{x-3}$. The limits at $\pm\infty$ equal 2. At 3^-, $f \to -\infty$; at 3^+, $f \to +\infty$. The asymptotes are $x = 3$ and $y = 2$.

Conjugation. For $x > 0$, $\sqrt{x^2+4x} - x = \frac{4}{\sqrt{1+\frac{4}{x}}+1} \to 2$. For $x < 0$, take care: $\sqrt{x^2} = |x| = -x$.

Exponential and comparison of growth rates

$$\lim_{x\to+\infty} e^x = +\infty, \quad \lim_{x\to-\infty} e^x = 0, \quad \lim_{x\to+\infty} \frac{e^x}{x^m} = +\infty \quad (m \in \mathbb{N}).$$

We deduce $x^m e^{-x} \to 0$ at $+\infty$.

Proof of the limits of the exponential. Studying $e^x - x - 1$ gives $e^x \geq 1 + x$. By comparison, $e^x \to +\infty$ at $+\infty$. Setting $t = -x \to +\infty$, $e^x = \frac{1}{e^t} \to 0$.

Proof of a comparison of growth rates. For every integer $k \geq 0$ and $x \geq 0$, $e^x \geq \frac{x^k}{k!}$. For $k = 0$, $e^x \geq 1$. If the inequality holds at order k, the derivative of $e^x - \frac{x^{k+1}}{(k+1)!}$ is $e^x - \frac{x^k}{k!} \geq 0$; its value at 0 is positive. The inductive step follows. Choosing $k = m + 1$, $\frac{e^x}{x^m} \geq \frac{x}{(m+1)!} \to +\infty$.

Choosing a transformation

Quotient of polynomials: factorise the highest powers. Difference of roots: conjugate. Quotient near a zero: factorise and study the signs. Exponential against power: comparison of growth rates, after a change of variable if necessary.

Comparison and composition: conditions of use

If $g(x) \leq f(x) \leq h(x)$ near the point or the infinity under study, and if g, h have the same real limit ℓ, then $f(x) \to \ell$ (squeeze theorem). If $g \leq f$ and $g \to +\infty$, then $f \to +\infty$. For a limit of $-\infty$, use an upper bound by a function tending to $-\infty$. If $u(x) \to a$ and v is continuous at a, then $v(u(x)) \to v(a)$, whenever the composite is defined. For example $\sqrt{1 + \frac{1}{x}} \to 1$ at $+\infty$. For an infinite limit of the outer function, check from which side $u(x)$ approaches the bound: $\frac{1}{u(x)}$ does not behave in the same way depending on the sign of $u(x)$.

Research activity

A curve close to a line — 30 min.

Consider the function f defined on $\mathbb{R} \setminus \{1\}$ by $f(x) = \frac{3x-2}{x-1}$. Let $\mathcal{C}_f$ be its curve in a plane frame of reference.

1. Show that, for every $x \neq 1$, $f(x) = 3 + \frac{1}{x-1}$.
2. Determine the limits of f at $-\infty$, at 1^-, at 1^+ and at $+\infty$.
3. Study the sign of $f(x) - 3$, then state the position of $\mathcal{C}_f$ relative to the line with equation $y = 3$.
4. Determine a real number $A > 1$ such that, for every $x > A$, $|f(x) - 3| < 0.01$.
5. Does the curve $\mathcal{C}_f$ cross its horizontal asymptote? Justify, and explain why a graphing window that is too wide can give a misleading impression.

Progressive exercises

E1 · Practice. Determine $\lim\limits_{x \to +\infty} \dfrac{5x^3 - x}{2x^3 + 7}$, with justification.

E2 · Practice. Let f be the function defined on $\mathbb{R} \setminus \{2\}$ by $f(x) = \frac{1}{(x-2)^2}$. Determine its limits at 2^-, at 2^+, at $-\infty$ and at $+\infty$.

E3 · Practice. Determine $\lim\limits_{x \to 1} \dfrac{x^2 - 1}{x - 1}$, stating why the simplification used is valid in a neighbourhood of 1.

E4 · Practice. Determine $\lim\limits_{x \to +\infty} x^3 e^{-2x}$. State the result on comparison of growth rates that is used.

E5 · Practice. Determine $\lim\limits_{x \to -\infty} \left(\sqrt{x^2 + 1} + x\right)$. Justify in particular the treatment of $\sqrt{x^2}$.

Textbook or past-paper work

Barbazo, p. 96, exercises 26 and 29. In S23: exercise 29, to link limits and asymptotes. At home after S24: exercise 26, to construct three counterexamples to an invalid rule for calculating with limits.

Homework

Homework 3 — A family of curves — 45 min.

For each real number a, consider the function f_a defined by $f_a(x) = \frac{ax^2+1}{x^2+1}$. Let $\mathcal{C}_a$ be its graph.

1. Determine the domain of f_a, then its limits at $-\infty$ and at $+\infty$.
2. Show that $f_a(x) - a = \frac{1-a}{x^2+1}$. Depending on the value of a, state the position of $\mathcal{C}_a$ relative to the line with equation $y = a$.
3. Study the variations of f_a by distinguishing the cases $a < 1$, $a = 1$ and $a > 1$. State any extrema and where they are attained.
4. Take $a = 3$. Solve, for $x \geq 0$, the inequality $|f_3(x) - 3| \leq 0.02$.

Teaching notes and solutions

Teacher notes

In S21, link each limit to a reading of the graph. In S23, require the signs of the denominator on the left and on the right. In S24, have students choose the transformation before calculating: factorisation, conjugation, local expansion not used. In S26, separate the proof of the comparison of growth rates from its use. Oblique asymptotes are not required here.

Solutions to the activity

1. Since $3x-2 = 3(x-1)+1$, for $x \neq 1$, $f(x) = 3 + \frac{1}{x-1}$.
2. At $\pm\infty$, $\frac{1}{x-1} \to 0$: the limits equal 3. At 1^-, $x-1 \to 0^-$, so $f(x) \to -\infty$; at 1^+, $x-1 \to 0^+$, so $f(x) \to +\infty$.
3. The sign of $f(x)-3 = \frac{1}{x-1}$ is negative for $x<1$, positive for $x>1$. The curve is therefore below and above the asymptote $y=3$ respectively.
4. For $x>1$, $|f(x)-3| < 0.01$ is equivalent to $\frac{1}{x-1} < 0.01$, then $x > 101$. We can choose $A = 101$.
5. No: $f(x)-3 = \frac{1}{x-1}$ is never zero. The curve may appear to coincide with the asymptote at the precision of the screen, but it never meets it.

E1. For $x \neq 0$, the quotient equals $\frac{5-\frac{1}{x^2}}{2+\frac{7}{x^3}}$. At $+\infty$, the inverse powers tend to 0 and the denominator to $2 \neq 0$. The limit is $\frac{5}{2}$.

E2. At 2, $(x-2)^2 \to 0$ through strictly positive values on both sides; its reciprocal therefore tends to $+\infty$. At $\pm\infty$, the square tends to $+\infty$ and its reciprocal to 0.

E3. For $x \neq 1$, $x^2-1 = (x-1)(x+1)$, so the quotient is equal to $x+1$. Its limit at 1 is therefore 2. This simplification calculates the limit; it does not make the original function defined at 1.

E4. Set $t = 2x \to +\infty$. Then $x^3e^{-2x} = \frac{t^3}{8e^t} \to 0$, by comparison of growth rates of the exponential and the power.

E5. The conjugate gives $\sqrt{x^2+1}+x = \frac{1}{\sqrt{x^2+1}-x}$. As $x \to -\infty$, both terms $\sqrt{x^2+1}$ and $-x$ tend to $+\infty$; the denominator therefore tends to $+\infty$. The limit is 0, through positive values.

Solutions to Homework 3

1. Since $x^2+1>0$ on $\mathbb{R}$, the domain is $\mathbb{R}$. Dividing by x^2 for $x \neq 0$, $f_a(x) = \frac{a+\frac{1}{x^2}}{1+\frac{1}{x^2}} \to a$ at $\pm\infty$.
2. $f_a(x)-a = \frac{ax^2+1-a(x^2+1)}{x^2+1} = \frac{1-a}{x^2+1}$. For $a<1$, the curve is above $y=a$; for $a>1$, it is below; for $a=1$, it coincides with this line.
3. By the quotient rule, $f_a'(x) = \frac{[2ax(x^2+1)-2x(ax^2+1)]}{(x^2+1)^2} = \frac{2x(a-1)}{(x^2+1)^2}$. If $a>1$, the sign is that of x: decreasing on $]-\infty,0]$, increasing on $[0,+\infty[$; minimum $f_a(0)=1$ and no maximum (the bound a is not attained). If $a<1$, the variations are reversed: maximum 1 and no minimum. For $a=1$, the function is constant, with value 1.
4. For $a=3$ and $x \geq 0$, the inequality becomes $\frac{2}{x^2+1} \leq 0.02$. The denominator is positive, so it is equivalent to $x^2+1 \geq 100$, that is $x \geq \sqrt{99}$.

Solutions to the textbook or past-paper work

Exercise 26. As $x \to +\infty$, the functions $f(x) = x$ and $g(x) = x$ both tend to $+\infty$, but their difference is zero. With $f(x) = 2x$, $g(x) = x$, the difference tends to $+\infty$; swapping f and g, it tends to $-\infty$. These examples establish that the difference is an indeterminate form.

Exercise 29. The denominator vanishes at $\frac{3}{2}$, so $D_f = \mathbb{R} \setminus \{\frac{3}{2}\}$. We write

$$f(x) = \frac{5 - 2x}{2x - 3} = -1 + \frac{2}{2x - 3}.$$

Thus $f(x) \to -1$ at both infinities. At $\frac{3}{2}$, the numerator of the second term is positive and its denominator tends to 0: the left-hand limit is $-\infty$, the right-hand limit is $+\infty$. The asymptotes have equations $y = -1$ and $x = \frac{3}{2}$.